

Mini UPS 12V AC-DC Power Backup System for Routers

I. Introduction

A stable internet connection is essential for online classes, remote work, communication, and entertainment. However, during power outages, routers immediately shut down, causing internet interruptions. This is especially common in the Philippines, where power interruptions frequently occur. Commercial UPS systems are often expensive or oversized for small devices such as routers. This project aims to design a compact and affordable UPS system that can continuously power routers during outages, ensuring uninterrupted internet connectivity.

II. Problem Statement

This project seeks to answer the following questions:

1. How can internet connectivity be maintained during power outages?
2. How can a low-cost UPS system be developed for routers?
3. How can rechargeable batteries provide reliable backup power?

III. Objectives

1. To develop a portable UPS system that provides backup power for routers during outages.
2. To maintain stable internet connectivity during emergencies or interruptions.
3. To reduce potential damage to routers caused by sudden power loss or voltage fluctuations.
4. To improve convenience for students, remote workers, and households relying on internet access.

IV. Scope and Limitations

Scope

- Designed for routers and modems
- Uses rechargeable batteries
- Provides temporary backup power during outages

Limitations

- Limited runtime depending on battery capacity
- Not designed for high-power appliances

V. Significance of the Study

This project helps students, workers, and households maintain internet access during unexpected power interruptions. By providing a low-cost and compact solution, it ensures that daily activities that depend on internet connectivity such as online learning, remote work, and communication remain unaffected during power outages.

VI. Related Systems

- Commercial mini UPS devices
- Power banks with DC output
- Traditional UPS systems

VII. Methodology

Planning Phase

During the planning phase, the Mini UPS system was first **simulated using circuit simulation software** to analyze its behavior before actual construction. The simulation included modeling the 18650 batteries, 3S Battery Management System (BMS), and boost converter to observe voltage output and current flow.

The goal of the simulation was to:

- Verify that the system could produce a stable 12V output required by the router
- Analyze charging and discharging behavior of the battery pack
- Identify potential issues such as voltage drops or instability
- Optimize component values and configuration before physical assembly

Results from the simulation helped ensure that the design would function correctly, reducing errors and improving efficiency during the actual building phase.

Figure 1. Voltage Source from Battery

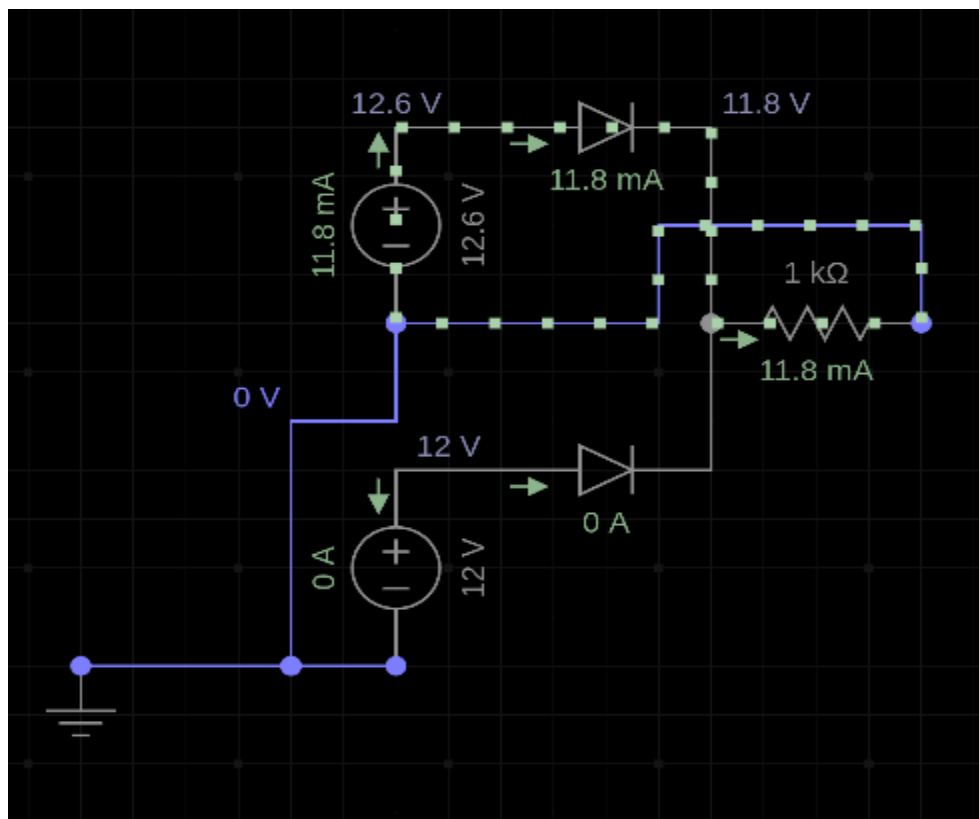
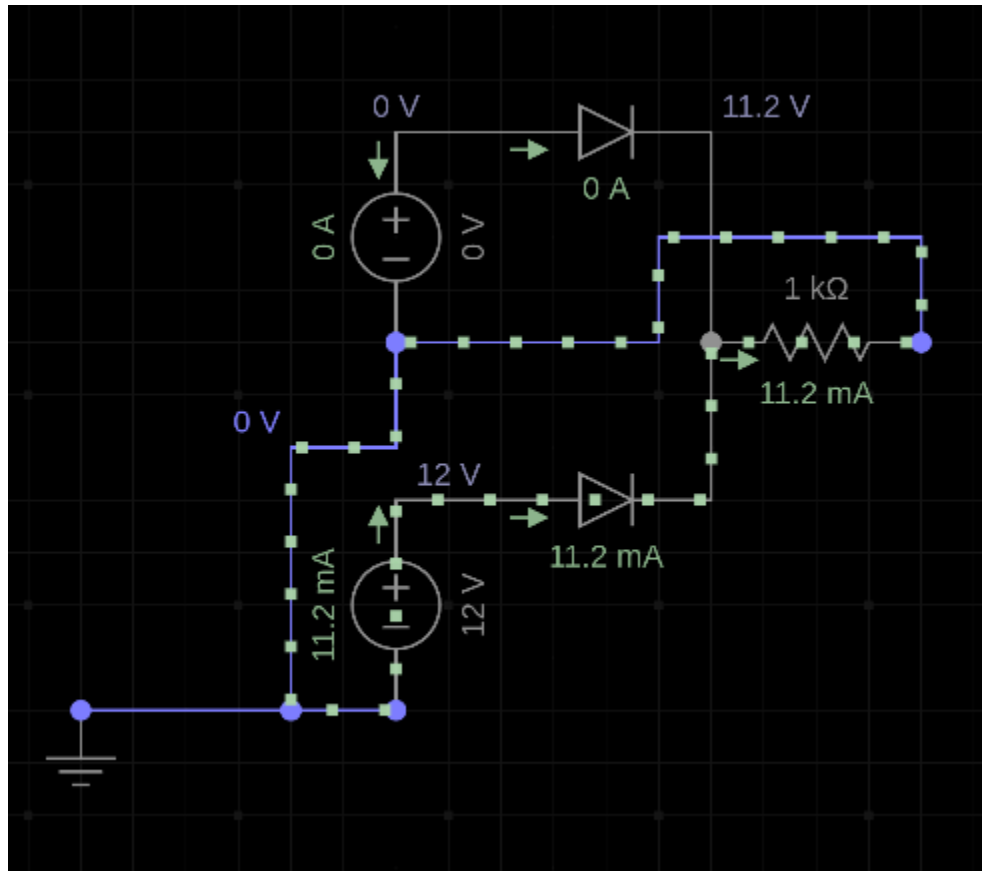


Figure 2. Voltage Source from Adapter

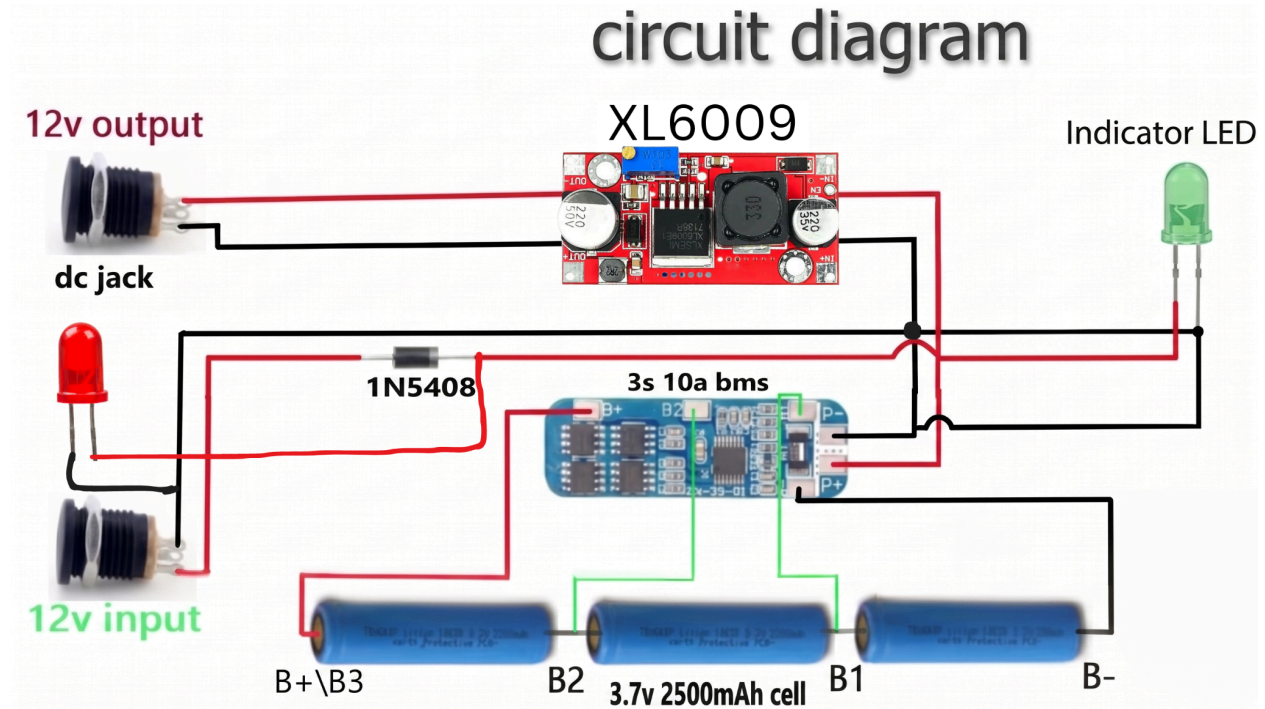


Design Phase

Design the UPS circuit using the following components:

- 18650 batteries
- 3S BMS (Battery Management System)
- Boost converter / voltage regulator
- Charging module

Figure 3. Schema



Building Phase

The components were assembled based on the circuit design. The three 18650 batteries were connected in series and linked to the 3S BMS for protection. The output was connected to a boost converter to maintain a stable 12V supply. All parts were soldered securely, insulated properly, and arranged in a compact setup.

Figure 4. Assembly

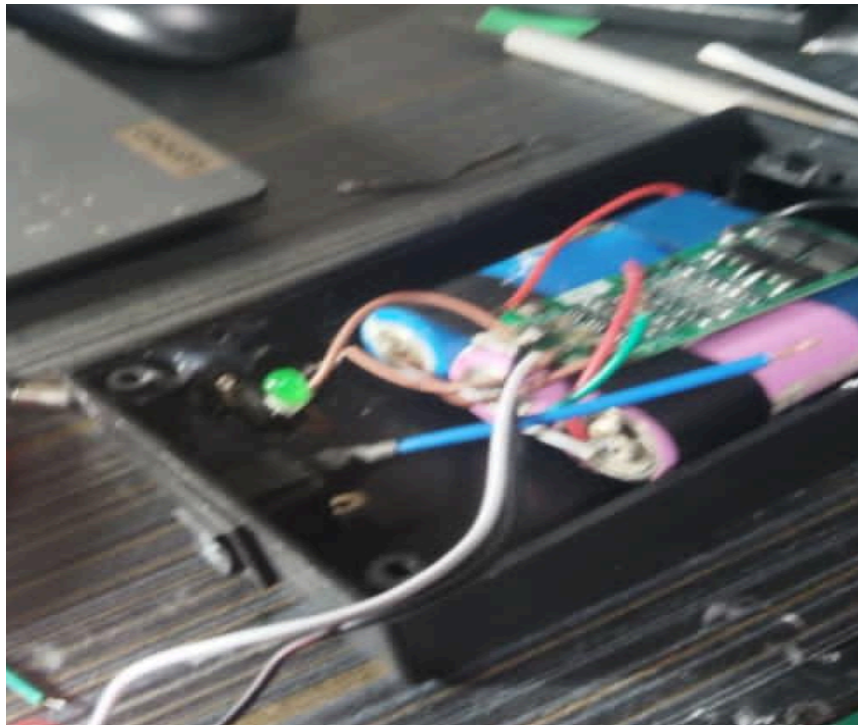
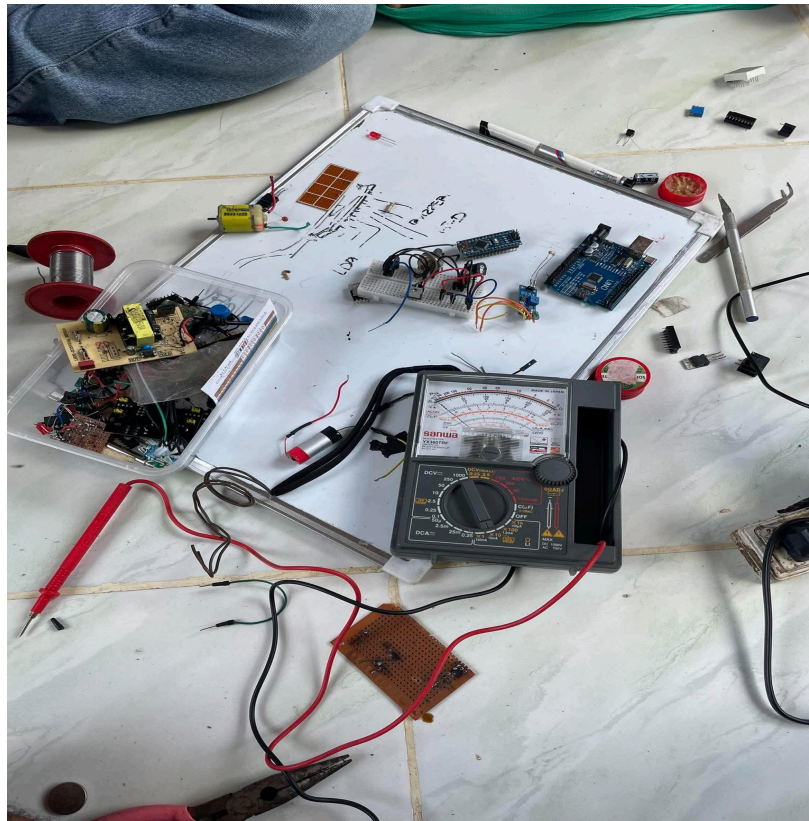


Figure 5. Assembly



Figure 6. Assembly



VIII. Materials Used

- 18650 lithium-ion batteries
- 3S Battery Management System (BMS)
- Boost converter (XL6009)
- Charging adapter
- Router (for testing)
- Wires and connectors
- Soldering materials
- Dc jacks
- Led red and black

IX. Final Output

The project successfully produced a fully functional Mini UPS system designed to provide backup power to a router during power outages. The system was able to deliver a stable 12V output, ensuring continuous internet connectivity without interruptions.

During testing, the Mini UPS demonstrated reliable performance, maintaining consistent voltage and safely powering the router for a reasonable duration depending on the battery capacity. The compact design also made the system portable and easy to use for household or small office applications.

Overall, the output met the project objectives by offering an efficient, low-cost, and practical solution for maintaining internet access during unexpected power interruptions.



X. Calculations

For batteries connected in series:

$$V_{total} = V_1 + V_2 + V_3 \quad V_{total} = V_1 + V_2 + V_3 \quad V_{total} = V_1 + V_2 + V_3$$

Each 18650 battery:

$$\text{Nominal voltage} = 3.7V$$

$$V_{total} = 3.7 + 3.7 + 3.7$$

$$V_{total} = 11.1V$$

Fully charged voltage:

$$4.2V \times 3 = 12.6V$$

Total Battery Energy:

$$E = V \times Ah$$

$$E = 11.1V \times 2.5Ah \quad E = 11.1V$$

$$E = 27.75Wh$$

$$\text{Total stored energy} = 27.75 \text{ watt} - \text{hours}$$

Estimated Backup Runtime:

$$t = \frac{\text{Battery Energy}}{\text{Load Power}}$$

$$t = \frac{27.75Wh}{12W}$$

$$t = 2.31 \text{ hours}$$

Estimated runtime:

≈ 2 hours and 18 minutes

Battery Input (Energy Stored)

Each 18650 battery:

Voltage = 3.7V (nominal)

Capacity = 2000 mAh

For 3 batteries in series (3S):

Total Voltage = $3.7V \times 3 = 11.1V$

Convert capacity to Ah:

2000 mAh = 2 Ah

$$E = V \times Ah$$

$$E = 11.1V \times 2Ah = 22.2Wh$$

Output Load (Router Consumption)

Typical router:

Voltage = **12V**

Current = **1A**

$$P = V \times I$$

$$P = 12V \times 1A = 12W$$

XI. Timeline

Activity	Week 1	Week 2	Week 3	Week 4
Research	✓			
Design		✓		
Assembly			✓	
Testing				✓

XII. Budget

Item	Estimated Cost
18650 Batteries 3x	P100
Battery Management System (BMS) 1x	P30
Boost Converter / Regulator 1x	P30
Miscellaneous (wires, connectors, soldering)	P50
Total	P210

XIII. Conclusion

The project provides an affordable and practical solution for maintaining internet connectivity during power outages through the development of a compact Mini UPS system specifically designed for routers. By utilizing readily available and cost-efficient components such as 18650 lithium-ion batteries, a Battery Management System (BMS), and a boost converter, the system is able to deliver stable and reliable power during interruptions.

The completed Mini UPS demonstrated consistent performance during testing, maintaining a steady 12V output and successfully powering a router for a reasonable duration depending on battery capacity. Its compact design also ensures portability and ease of use, making it suitable for everyday applications in homes, small offices, and academic environments.

Furthermore, the project highlights the effectiveness of integrating basic electrical and electronic principles into a functional real-world application. It serves as a viable alternative to commercially available UPS devices, which are often more expensive and less accessible for basic needs.

Overall, the system proves to be a reliable, efficient, and cost-effective solution that can significantly benefit students, remote workers, and households across the Philippines by ensuring uninterrupted internet access during unexpected power outages.